Saurav Kumar

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Education

- **Technische Universität Darmstadt** — Darmstadt, Germany
  Semester Exchange Student; CGPA: 8.46/10 — *Oct 2025 – Mar 2026*
  - **Relevant Coursework**: Artificial Intelligence, Computer Vision, Probabilistic Graphical Models, LLMs, Scientific Computing
- **Indian Institute of Technology, Mandi** — Mandi, Himachal Pradesh, India
  B.Tech in Bioengineering; CGPA: 7.97/10 — *Aug 2023 – Present*
  - **Relevant Coursework**: Data Structures & Algorithms, Machine Learning, Python & Data Science, Probability & Statistics

Technical Skills

Languages: Python, C++, JavaScript, Perl | **Frameworks**: React, ROS, TensorFlow, Django, Streamlit

Tools: Git, Linux, Docker, SolidWorks, Raspberry Pi, Arduino | **Concepts**: Sensor Fusion, Kalman Filtering, Control Systems

Experience

- **Morsebiz** — Remote
  Web Development Intern — *Jan 2026 – Mar 2026*
  - **AI Data Processing**: Developed high-throughput web scraping pipelines coupled with an AI-driven content verification system, achieving 98% accuracy in content verification across 10,000+ daily data entries.
  - **System Integration**: Engineered automated communication workflows by integrating asynchronous messaging APIs, enabling real-time system alerts and seamless cross-platform user engagement.
  - **Automated Billing Ops**: Architected a dynamic, automated invoice generation module that streamlined billing operations, reducing manual processing time by 40% and ensuring strict transactional compliance.

Projects

- **Predicting Protein Stability Changes ($\Delta\Delta G$)** — Live | Paper
  Machine Learning, Siamese GNN, ESM-2, Bayesian Networks — *Feb 2026 – May 2026*
  - **Architecture Design**: Engineered a 1D-3D latent fusion architecture combining a Siamese Graph Neural Network (GNN) with a 150-million parameter Protein Language Model (ESM-2).
  - **Predictive Performance**: Achieved a Pearson Correlation of 0.7266 and Mean Absolute Error of 0.7472 kcal/mol, significantly surpassing traditional biophysical force fields.
  - **Causal Inference**: Constructed a Hybrid Bayesian Network to map explicit biological causality, mathematically quantifying the vulnerability of protein surfaces to polarity shifts.
- **Vaya – Your Local Healthcare Connection** — GitHub
  Python, Django, Django Channels, Groq API, RAG — *Aug 2025 – Mar 2026*
  - **Full-stack Platform**: Designed doctor discovery, appointment booking, and report handling system with secure authentication.
  - **AI RAG System**: Implemented a two-step retrieval-augmented generation pipeline, achieving 85% answer accuracy and reducing hallucination rate by 30% compared to baseline models.
  - **Real-time Messaging**: Built encrypted, real-time patient-doctor chat with Django Channels.
- **AquaSweep – Underwater Rover** — GitHub
  Python, OpenCV, Flask, ROS, Arduino, Raspberry Pi — *Jan 2025 – Jun 2025*
  - **Live Streaming**: Engineered MJPEG camera pipeline (20 FPS, < 200 ms latency) using Flask sockets.
  - **Edge AI Detection**: Built OpenCV pipelines for algae and motion detection with ~93% lab accuracy.
  - **Depth Stabilization**: Controlled 8 thrusters via ROS–Arduino; fused IMU and Bar30 data with Kalman filtering (±3 cm error).

Leadership & Achievements

Awards & Honors: Selected for TU Darmstadt semester exchange (Top 450+ batch merit) | Secured Top 5 rank in Arduino Hackathon.

Extracurriculars: Hospitality Head for Exodia (managed 1000+ participants) | Robotics Club Mentor, IIT Mandi.